

CONGLING TANG

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EDUCATION

Duke University <i>Master of Engineering, Robotics & Autonomy</i>	Aug. 2026 – Dec. 2027 (Expected) Durham, NC
McMaster University <i>Bachelor of Engineering, Mechatronics Engineering (GPA: 3.8/4.0)</i>	Sep. 2021 – Jun. 2026 Hamilton, ON

WORK EXPERIENCE

Duke University <i>Teaching Assistant, Undergraduate Control Systems</i> Led weekly office hour sessions and graded coursework on feedback control, PID tuning, and state-space analysis	Aug. 2026 – Present Durham, NC
Husky Technologies <i>Platform Control Software Intern</i> - Engineered a machine simulation framework using Beckhoff's TE1111 module and ADS, covering full-motion and servo driver simulation across machine configurations - Developed an auto-generation C# tool to automate TwinCAT PLC/TSM project creation, integrating APIs into production via Ruby - Designed and implemented new HMI pages using domain-specific scripting to streamline operator workflows - Streamlined project lifecycles and build configurations in Team Foundation Server (TFS) for version control	May 2024 – Aug. 2025 Bolton, ON

RESEARCH EXPERIENCE

McMaster University <i>Research Course – Magnetometer-Based Orientation Estimation</i> - Developed a MATLAB framework for 3D orientation estimation with an Extended Kalman Filter; simulated magnetometer data and noise to model and validate state estimation - Modeled Microwave-Assisted Optical Pumping physics in Python to characterize hyperfine structure and Zeeman spectra	Sep. 2025 – Dec. 2025 Hamilton, ON
McMaster University <i>Research Assistant – NTIRE Workshop (CVPRW)</i> Co-Author: "SwinFSR: Stereo Image Super-Resolution using SwinIR and Frequency Domain Knowledge" - Executed validation pipeline: curated datasets, built training loops, and tuned hyperparameters for super-resolution	Feb. 2023 – Apr. 2023 Hamilton, ON

PROJECTS

Flexible Robotic Production System (F.R.P.S.) <i>Honda of Canada Mfg. / McMaster Capstone</i> - Developed an autonomous pipeline for optimal motion planning and collision avoidance using ROS2 and MoveIt2 - Integrated stereo-camera perception for hand-eye calibration in simulation; planned physical deployment via Phidget boards	Aug. 2025 – Jul. 2026 Hamilton, ON
DIY 3D-LiDAR Scanner <i>Personal Project</i> - Designed and 3D-printed a custom nodding mechanism in Fusion 360 to repurpose a 2D LiDAR into a 3D LiDAR using a stepper motor - Integrated motor encoder feedback with LiDAR range data to generate and visualize real-time 3D point clouds in RViz2	Jun. 2026 Hamilton, ON
Mac RoboMaster <i>Computer Vision Team Lead</i> - Trained YOLO models with custom datasets and OpenCV for real-time auto-aiming; built state-machine target tracking with Kalman Filters - Deployed 3D LiDAR-based navigation and obstacle avoidance on Jetson Orin using ROS2 Nav2	Oct. 2022 – Sep. 2025 Hamilton, ON
Autonomous Sumobot <i>1st Place Winner, McMaster Sumobot Competition</i> Architected detection algorithms and sensor layouts using IR/ultrasonic arrays for autonomous combat logic on Arduino	Sep. 2023 – Jan. 2024 Hamilton, ON
NASA Air-LUSI Equipment Housing <i>Research Assistant, McMaster ICE Lab</i> - Fabricated a customized force sensor and modeled high-precision ball screws to support laboratory calibration - Designed protective mechanical housing components for the NASA Air-LUSI Airborne Lunar Spectrometer	Jun. 2023 Hamilton, ON

TECHNICAL SKILLS

Languages: C++, Python, C, C#, PLC, MATLAB, Simulink, Ruby
Robotics & Frameworks: ROS/ROS2, MoveIt2, Nav2, OpenCV, YOLO, Gazebo, Beckhoff TwinCAT
Hardware & Embedded: STM32, Arduino, Raspberry Pi, Jetson Orin/Nano, 3D Printing, Soldering
CAD & Tools: Fusion 360, SolidWorks, Inventor, Git, Docker, Keil